



FDE WEEK 1

Developer Setup Playbook

Windows + macOS

Goal

Complete this setup before the first weekend session so classroom time can be spent coding, not installing tools. Expected setup time: 30-60 minutes.

Install

- VS Code Insiders
- Python 3.13.x
- Python extension
- Claude Code extension

Configure

- Create .venv
- Select Python interpreter
- Install Week 1 packages
- Run Python + FastAPI pre-flight

Week 1 scope: Python foundations → file/data handling → exception handling → FastAPI basics

1. What you need before you start

Recommended Python version

Use Python 3.13.x for this cohort. Python 3.14 is current, but keeping everyone on the same 3.13.x line reduces avoidable compatibility differences during the first few weeks.

Official downloads

VS Code Insiders: <https://code.visualstudio.com/insiders/>

Python for Windows: <https://www.python.org/downloads/windows/>

Python for macOS: <https://www.python.org/downloads/macos/>

Claude Code in VS Code: <https://docs.anthropic.com/en/docs/claude-code/ide-integrations>

Required VS Code extensions

- Python — Microsoft
- Pylance — Microsoft (recommended for IntelliSense/type support)
- Claude Code — Anthropic
- Optional: Jupyter — only if you want notebook-based experimentation later

Claude account requirement

Claude Code in VS Code requires an eligible Anthropic account (for example a supported paid Claude subscription or Claude Console account).

Create your Week 1 workspace

```
fde-week1/  
├── .venv/  
├── data/  
│   └── sample.csv  
├── src/  
│   ├── hello.py  
│   └── app.py  
└── requirements.txt
```

2. Windows setup

STEP	ACTION	WHAT TO DO
1	Install VS Code Insiders	Download the Windows 64-bit build from the official VS Code Insiders page. Install using the default options.
2	Install Python	Install Python 3.13.x (64-bit) from python.org. Use the standard installer / Python install manager available for your machine.
3	Open VS Code Insiders	Launch VS Code Insiders and open the Extensions view using Ctrl+Shift+X.
4	Install extensions	Install Python (Microsoft), Pylance (Microsoft) and Claude Code (Anthropic).
5	Create project folder	Create a folder named fde-week1 and open that folder in VS Code Insiders.
6	Create virtual environment	Open Terminal inside VS Code and create a .venv using the command below. (Page 4)
7	Select interpreter	Ctrl+Shift+P → “Python: Select Interpreter” → choose the interpreter inside .venv.
8	Install packages	Activate the virtual environment, upgrade pip and install the Week 1 packages.
9	Sign in to Claude Code	Open the Claude Code panel and sign in with your Anthropic / Testleaf-provided account.

Windows commands

```
# Verify Python
py -3.13 --version

# Create virtual environment
py -3.13 -m venv .venv

# Activate - PowerShell
.\venv\Scripts\Activate.ps1

# OR activate - Command Prompt
.venv\Scripts\activate.bat

# Upgrade pip
python -m pip install --upgrade pip

# Install Week 1 packages
pip install requests fastapi "uvicorn[standard]"
```

If PowerShell blocks activation

Do not spend class time changing system-wide security settings. Use Command Prompt activation (.venv\Scripts\activate.bat) or simply select the .venv interpreter in VS Code. Ask the mentor if your corporate laptop enforces PowerShell restrictions.

Windows validation

```
python --version
pip --version
python -c "import requests, fastapi, uvicorn; print('Week 1 packages OK')"
```

3. macOS setup

STEP	ACTION	WHAT TO DO
1	Install VS Code Insiders	Download the Mac Universal build from the official VS Code Insiders page and move it to Applications.
2	Install Python	Install Python 3.13.x using the official macOS installer from python.org.
3	Open VS Code Insiders	Launch VS Code Insiders and open Extensions using Cmd+Shift+X.
4	Install extensions	Install Python (Microsoft), Pylance (Microsoft) and Claude Code (Anthropic).
5	Create project folder	Create a folder named fde-week1 and open that folder in VS Code Insiders.
6	Create virtual environment	Open Terminal inside VS Code and create a .venv using python3.13 using the commands below.(page 6)
7	Select interpreter	Cmd+Shift+P → “Python: Select Interpreter” → choose .venv/bin/python.
8	Install packages	Activate .venv, upgrade pip and install the Week 1 packages.
9	Sign in to Claude Code	Open the Claude Code panel and sign in with your Anthropic.

macOS commands

```
# Verify Python
python3.13 --version

# Create virtual environment
python3.13 -m venv .venv

# Activate
source .venv/bin/activate

# Upgrade pip
python -m pip install --upgrade pip

# Install Week 1 packages
pip install requests fastapi "uvicorn[standard]"
```

Important on macOS

Use the Python installed for this course rather than relying on any system-provided Python. After activating `.venv`, the command “which python” should point inside your `fde-week1/.venv` folder.

macOS validation

```
python --version
which python
pip --version
python -c "import requests, fastapi, uvicorn; print('Week 1 packages OK')"
```

4. Pre-flight check: Python

Create `src/hello.py` with the following code:

```
name = "FDE Learner"
skills = ["Python", "APIs", "AI"]

for skill in skills:
    print(f"{name} is getting ready for {skill}")
```

Run:

```
python src/hello.py
```

Expected result

You should see three lines printed in the terminal, one each for Python, APIs and AI. If this works, VS Code + Python + your virtual environment are ready.

5. Pre-flight check: FastAPI

Create `src/app.py` with:

```
from fastapi import FastAPI

app = FastAPI(title="FDE Week 1 API")

@app.get("/health")
def health():
    return {"status": "ready", "week": 1}
```

Start the API from the project root:

```
uvicorn src.app:app --reload
```

Open in your browser: <http://127.0.0.1:8000/docs>

Expected result

Swagger UI should open and show GET /health. Use "Try it out" → Execute.

A successful response should return status = ready and week = 1.

6. Ready-for-class checklist

- ☐ VS Code Insiders launches successfully.
- ☐ Python 3.13.x is installed.
- ☐ The fde-week1 project folder opens in VS Code.
- ☐ .venv has been created.
- ☐ VS Code is using the .venv Python interpreter.
- ☐ requests, FastAPI and Uvicorn are installed inside .venv.
- ☐ hello.py runs successfully.
- ☐ FastAPI Swagger opens at /docs.
- ☐ Claude Code extension is installed and you can open/sign in to it.

Quick troubleshooting

ISSUE	WHAT TO TRY
python / py not found	Close and reopen VS Code after Python installation. On Windows try “py -3.13”. On Mac try “python3.13”.
Wrong interpreter	Command Palette → Python: Select Interpreter → select the interpreter inside .venv.
pip installs globally	Activate .venv first, then run “python -m pip ...”. Verify “python” points to .venv.
PowerShell activation blocked	Use Command Prompt activation or select the .venv interpreter directly in VS Code.
Port 8000 already used	Run: uvicorn src.app:app --reload --port 8001
Claude panel not visible	Restart VS Code Insiders or run Developer: Reload Window. Search “Claude Code” in the Command Palette.